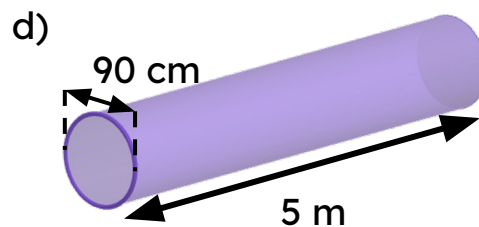
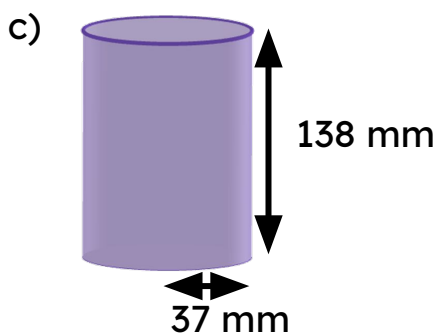
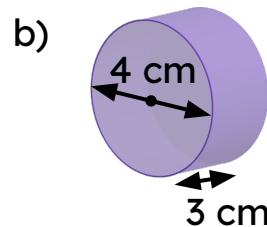
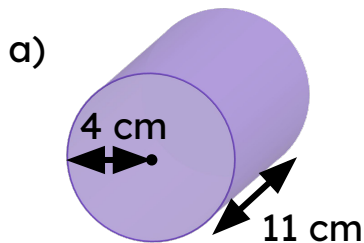




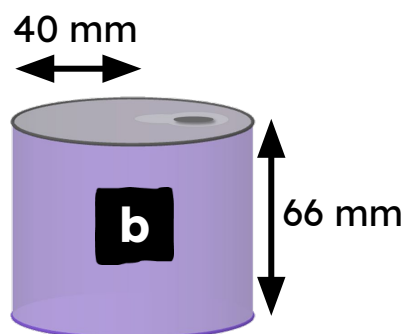
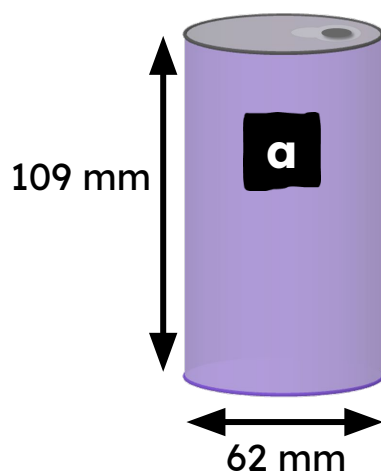
Surface area of cylinders

Task A: Finding the surface area of a cylinder

1) Work out the surface area of the following **cylinders**

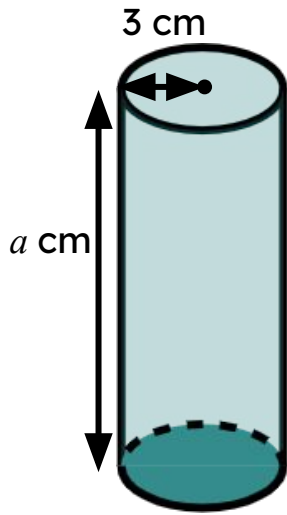


2) A drinks company are looking to use less material when making their cans. These two cans both hold 330 ml of liquid. Which can should the drinks company choose?

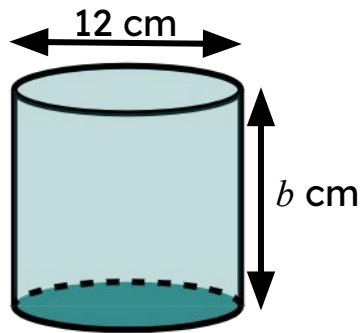


Task B: Using the surface area of a cylinder

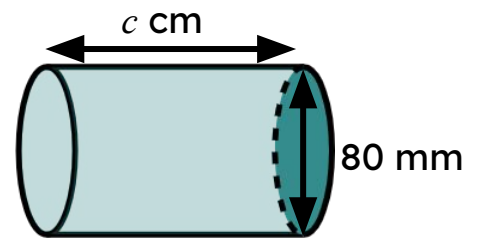
1) Find the missing length in each question. Give your answer in cm.



Surface area
 $= 78\pi \text{ cm}^2$



Surface area
 $= 168\pi \text{ cm}^2$



Surface area
 $= 88\pi \text{ cm}^2$

2) A necklace is being made. Part of the design is a metal cylindrical rod.

The rod is gold-plated. $208\pi \text{ mm}^2$ of gold leaf is used. The rod has a diameter of 4 mm.

How long is the rod?